PROJECT PHASE-I REPORT

DEEP LEARNING-BASED REHABILITATION SYSTEM FOR POST-STROKE MOTOR FUNCTION RECOVERY

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by

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Certificate

This is to certify that the project Phase-I report titled **DEEP LEARNING-BASED REHABILITATION SYSTEM FOR POST-STROKE MOTOR FUNCTION RECOVERY** is a bonafide record of the work carried out by **Anagha K.R (VAS23EC018), Anitta Johnson (VAS23EC022), Devika T.S (VAS23EC035) and Franil Shaju P (VAS23EC037)** of Vidya Academy of Science & Technology, Thrissur, in partial fulfilment of the requirements for the award of the **Degree of Bachelor of Technology** in **Electronics and Communication Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2026-2027. The report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the said degree.

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PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

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PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

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During the course of our Phase-I project work, several persons collaborated directly and indirectly with us. Without their support, it would have been impossible for us to finish our work. That is why We wish to dedicate this section to recognise their support.

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ABSTRACT

Stroke is one of the leading causes of long-term disability, and the recovery of lost motor function depends on regular, correctly performed rehabilitation exercises. Conventional rehabilitation needs continuous therapist supervision, which is costly and inconvenient, while unsupervised home exercise gives no monitoring and no feedback. Wearable-sensor systems and rehabilitation robots are expensive and tie the patient to special hardware.

This project proposes a low-cost, camera-based system that uses computer vision and deep learning to monitor post-stroke upper-limb exercises. A webcam captures the patient while exercising, the frames are pre-processed, and a pose-estimation module (MediaPipe, with YOLO11-Pose for comparison) extracts the body keypoints. Joint angle, range of motion, speed and smoothness are computed from the keypoints and given to deep learning models (CNN, LSTM/GRU, hybrid CNN–LSTM, and a GAN for data augmentation) that recognise the exercise and judge the quality of the movement. A weighted score between 0 and 100 is calculated, real-time visual and audio corrective feedback is given, and the results are stored in a database so that daily and weekly progress reports can be produced for the patient and the therapist.

In Semester VII the problem was defined, the literature was surveyed, the architecture, database and screens were designed and the tools were selected. Reference code for feature extraction, repetition counting and scoring was run on a synthetic test signal and returned three repetitions and a range of motion of 80°, as expected. The comparison plots of the candidate algorithms are illustrative only and will be replaced by measured results. Data collection and the pose-estimation module are in progress; training, integration and testing with users will be completed in Semester VIII.

Keywords: stroke rehabilitation, computer vision, deep learning, pose estimation, LSTM, home-based rehabilitation

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LIST OF SYMBOLS AND ABBREVIATIONS

$\mathbf{p}_{j,t}$	Keypoint position of joint j at frame t
ADL	Activities of Daily Living
AP	Average Precision
ARAT	Action Research Arm Test
CNN	Convolutional Neural Network
COCO	Common Objects in Context (keypoint format)
CV	Computer Vision
DL	Deep Learning
EMG	Electromyography
FMA	Fugl-Meyer Assessment
FPS	Frames Per Second
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
HPE	Human Pose Estimation
J	Normalised jerk (smoothness measure)
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
mAP	mean Average Precision
ML	Machine Learning
ROM	Range of Motion
S	Performance score (0–100)
s_k, w_k	Sub-score k and its weight
SQL	Structured Query Language
T	Number of frames in one input window
θ_B	Angle at joint B
v_t	Movement speed at frame t
VR/AR	Virtual Reality / Augmented Reality
YOLO	You Only Look Once

Chapter 1

INTRODUCTION

A stroke is usually described as a sudden event, but for the person who survives it, the real story begins afterwards. Within minutes, a part of the brain loses its blood supply, either because a vessel is blocked (ischaemic stroke) or because a vessel bursts (haemorrhagic stroke). The nerve cells in that region stop working, and the arm, the leg or the face on the opposite side of the body may become weak or stop obeying the person's wishes. Stroke is one of the leading causes of long-term disability in the world [5], and a large part of the disability is a loss of motor function. This chapter explains the problem, why it matters, what is planned to be done about it and how the rest of the report is arranged.

1.1 Background

The brain is not a fixed machine. After an injury, the cells that survive can form new connections and take over some of the work of the damaged region. This ability is called neuroplasticity. Studies of experience-dependent plasticity show that it is driven by repetition: the brain reorganises when a task is practised many times, with enough intensity, and when the practice is done correctly [6]. For this reason, rehabilitation exercises are the most important part of recovery after a stroke. Medicines and surgery save life and limit the damage, but it is the daily practice of movements such as raising the arm, bending the elbow or opening the hand that gives back the ability to eat, dress and write.

In a hospital or a rehabilitation centre, the exercises are guided by a physiotherapist or an occupational therapist. The therapist shows the movement, watches the patient perform it, corrects mistakes such as leaning the trunk instead of lifting the arm, counts the repetitions and judges whether the patient is improving. The assessment is usually done by observation and with clinical scales such as the Fugl-Meyer Assessment [7]. This model works well, but it needs the therapist to be present for every session. Frequent hospital visits are costly and tiring, and they are especially difficult for elderly patients, for patients who live far from a centre and for families with a single earner.

After the first few weeks, most patients are therefore asked to continue the exercises at home. At home, nobody checks whether the movement is correct, whether the range of motion is growing from week to week or whether the patient has slipped into a compensating movement that gives the feeling of progress but not the real benefit. Many patients also lose motivation because they do not see any measurable result. The therapist, on the other hand, sees the patient only once in a few days or weeks and has to depend on the patient's own account.

At the same time, three things have become cheap and easy. Ordinary cameras are

available in every laptop and mobile phone. Pose-estimation software can find the position of the shoulders, elbows and wrists in a video in real time without any special sensor [9, 10]. And deep learning has shown that neural networks can learn to recognise and judge human movement from sequences of such body points [1, 4]. This project brings these three things together into a camera-based rehabilitation assistant that watches the exercise, tells the patient how to correct it, and keeps a record of progress for the patient and the therapist (Figure 1.1).

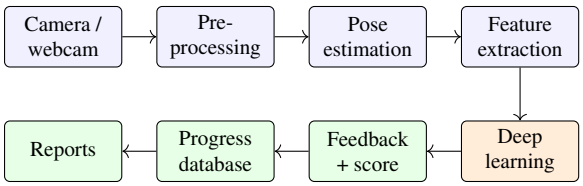


Figure 1.1: Proposed vision-based rehabilitation system (overview)

1.2 Problem Definition

The problems that motivate this project can be stated as follows:

- Stroke patients often suffer from impaired motor functions that need long-term, regular rehabilitation.
- Traditional rehabilitation requires continuous therapist supervision, which is costly and limited by the number of available therapists.
- Home-based rehabilitation lacks proper monitoring and feedback, so mistakes go unnoticed and progress is not measured.
- Manual assessment of recovery is time-consuming and subjective; two therapists may judge the same movement differently.
- There is a need for an automated system that analyses patient movements using computer vision and deep learning and that can be used at home with simple equipment.

1.3 Objectives of the Work

The main aim is to build a low-cost, camera-based system that supervises post-stroke rehabilitation exercises, corrects the patient in real time and records recovery progress. The specific objectives are:

1. To develop an AI-based rehabilitation system for post-stroke patients.
2. To capture and analyse patient movements using a camera.
3. To detect body posture through computer vision techniques.
4. To apply deep learning for exercise recognition and assessment.
5. To provide real-time feedback and corrective guidance.
6. To monitor and track patient recovery progress.

7. To generate performance reports for therapists and patients.
8. To support home-based rehabilitation programmes.

To reach these objectives the following tasks have been set:

- study the existing rehabilitation methods and the vision-based assessment work, and note where they fall short;
- choose the hardware (camera and computer) and the software tools (pose estimation, deep learning framework, database and user interface);
- collect or select a dataset of upper-limb rehabilitation movements and prepare it for training;
- implement the pose-estimation and feature-extraction modules;
- design, train and compare deep learning models for exercise recognition and quality assessment;
- build the real-time feedback screen, the progress-tracking module and the report generator;
- test the complete system with different users, and evaluate its accuracy and response time.

1.4 Motivation for the Work

The purpose of the project is to help post-stroke patients recover their motor functions effectively by monitoring their rehabilitation exercises automatically. The motivation comes from the gap between the care that is needed and the care that is available:

- **A growing burden.** The number of stroke survivors who need long-term rehabilitation is large, while the number of therapists is limited. A system that handles routine supervision can free the therapist for the cases that need expert attention.
- **Cost and distance.** Travelling to a centre several times a week costs money and a day of work for the patient or a family member. A camera at home costs very little in comparison.
- **Lack of feedback at home.** Exercises that are done wrongly give less benefit and may even cause pain. Immediate, correct feedback is the main thing missing from home-based rehabilitation.
- **Objective measurement.** Numbers such as the angle reached by the elbow, the speed of the hand and the smoothness of the movement are more reliable than the impression of a single observer, and they can be compared from week to week.
- **A chance to apply our subject.** The project joins signal and image processing, machine learning, embedded and software systems, which are all part of the Electronics and Communication curriculum, and applies them to a real human need.

1.4.1 Social and Ethical Relevance

The project is meant for people who are weak, often elderly and sometimes anxious about technology. A system for such users has to be simple to operate, patient in its instructions and safe. Four rules are kept in mind throughout the design.

First, the system is an *aid* and not a replacement for the therapist. The exercises and the target ranges are set by the therapist, and the report is meant to help the therapist decide, not to decide for the therapist. Second, the privacy of the patient must be protected. A video of a person at home is sensitive. In the design, the video is processed on the patient's own computer, only the keypoints and the scores are stored, and the raw video is not saved unless the patient agrees. Third, the patient's safety must come first: the system should never push the patient beyond the range set by the therapist, and it should stop and ask the patient to rest when the movement becomes unstable. Finally, any trial with real patients will need proper consent and the approval of the college and the hospital before it is started.

1.4.2 Scope and Limitations

The system is designed for *upper-limb* rehabilitation exercises, namely shoulder, elbow and wrist movements, because the upper limb is the part for which a front-facing camera gives the clearest view. It uses one camera and works while the patient is sitting or standing in front of it. It is intended for use at home or in a clinic under the guidance of a therapist.

The system also has limits that are stated clearly. Its accuracy depends on good lighting, a clear view of the patient and the quality of the pose-estimation model. Two-dimensional keypoints from a single camera cannot measure movements that go straight towards or away from the camera as accurately as movements across the image. The deep learning models will be trained on the data that can be collected or obtained from public sources, and the models should not be considered clinically validated until they are tested with patients under medical supervision. Finally, the score produced by the system is not a clinical diagnosis.

1.5 Methodology Adopted

A pipeline approach is followed, shown in simplified form in Figure 1.1. A camera records the patient performing a prescribed exercise. The video is pre-processed (frame extraction, noise removal and enhancement), and a pose-estimation model detects the body keypoints in every frame. From the keypoint trajectories the joint angles, the range of motion and the movement speed are calculated. These features are given to deep learning models (CNN, LSTM/GRU, hybrid models and a YOLO-based pose model are being compared), which learn the pattern of correct and incorrect movement. The assessment module converts the output into a performance score and a recovery estimate. The system gives visual and audio feedback in real time, generates daily and weekly reports and stores all records in a database. The tools used are Python, OpenCV, MediaPipe, TensorFlow/PyTorch and SQLite; the details are given in the later chapters.

The work is divided into the following phases, which follow the timeline of the project (Figure 5.1 in Chapter 5):

1. **Problem identification and literature survey** (June–July 2026): choosing the title, studying the existing work and fixing the objectives.
2. **Dataset collection and system design** (July–August 2026): studying the available datasets, collecting rehabilitation videos and designing the system architecture.
3. **Pose estimation and motion analysis** (August–September 2026): implementing the keypoint extraction and the motion-analysis module.
4. **Deep learning model development** (September–October 2026): designing, training and validating the models.
5. **System integration and real-time feedback** (October–November 2026): joining the modules and building the user interface.
6. **Testing, evaluation and optimisation** (November–December 2026): testing with different users and improving the system.
7. **Documentation, final testing and submission**: preparing the report, the paper and the final demonstration.

1.6 Outline of the Report

This report contains six chapters. Chapter 1 gives the introduction to the project work and describes the problem, the objectives, the motivation, the scope and the methodology. Chapter 2 presents the literature review, the existing rehabilitation systems and the research gap. Chapter 3 explains the system design and methodology: the block diagram, pose estimation, feature extraction, the deep learning models, the assessment and feedback logic and the performance metrics. Chapter 4 describes the implementation: the software modules, the session flow, the database, the screens and the set-up. Chapter 5 presents the work completed in Semester VII and discusses the preliminary results. Chapter 6 gives the conclusion, the work plan for Semester VIII, the risks, the future scope of the work, and the mapping of the project with the POs, PSOs and SDGs. The reference code sketches are given in the Appendix.

1.7 Summary

This chapter introduced post-stroke rehabilitation and explained why supervised exercise is difficult to provide in sufficient quantity, while unsupervised exercise at home gives no feedback and no measurement. The problem, the objectives, the motivation, the scope and the limits of the work were stated, and the pipeline approach that has been chosen was described. The next chapter reviews what other researchers have done in this field.

Chapter 2

LITERATURE REVIEW

Before designing anything, it was necessary to know how others have used computer vision and deep learning to look at the movement of stroke patients, which datasets and tools exist, and which problems are still open. Four recent papers were read closely, and some basic references on clinical assessment, pose estimation and neural networks were studied to understand the background. This chapter describes the four papers, the background studies, the existing rehabilitation systems and the gap that the project tries to fill.

2.1 Paper I: Skeleton–Optical Flow Fusion for Post-Stroke Movement Analysis

Han, Wang, Li and Hu [1] developed a computer vision and deep learning framework that tells post-stroke patients apart from healthy individuals using ordinary video. Their method fuses two complementary kinds of motion information. The first is the *skeleton*, the set of body keypoints that describe posture and joint movement. The second is *optical flow*, which describes the apparent motion of pixels between consecutive frames and so captures fine movement that the keypoints may miss [14].

The work shows that a camera alone, without any wearable sensor, carries enough information to separate impaired movement from normal movement. For this project the paper supports the choice of skeleton keypoints as the main input of the learning model. It also suggests that optical flow could be added later as a second input stream if the keypoints alone do not give enough accuracy.

2.2 Paper II: Stroke Vision Bench

Robinson, Gupta, Qureshi, Fu and Shah [2] introduced a benchmark that contains stroke rehabilitation videos together with 2D skeletal pose data, meant for the objective assessment of upper-limb recovery with computer vision. Deep learning needs a large amount of labelled data, and one of the main obstacles in stroke research is that such data are hard to collect, because patients are few at any one centre and their videos are sensitive. A public benchmark of this kind is therefore valuable, because different methods can be compared on the same data.

This benchmark is considered a candidate source of training and validation data for the project, and its format and licence will be studied before it is used. Even if it is used,

a small set of own videos will still be recorded so that the system can be tested in the kind of home environment where it will actually be used.

2.3 Paper III: Using MediaPipe to Track Upper-Limb Reaching After Stroke

Wagh, Scott, Andrushko and co-authors [3] used MediaPipe to track upper-limb reaching movements after stroke with a single camera, for the purpose of rehabilitation assessment. MediaPipe is a free framework that gives body landmarks from normal video and runs fast enough on a laptop or even a phone [10, 11]. The study shows that this lightweight, freely available tool can be used to follow and measure reaching movements, which is exactly the low-cost, single-camera set-up that this project aims for.

MediaPipe is therefore one of the primary pose-estimation tools in the design. The paper also reminds us that landmarks from a single camera have limits, for example when the arm moves towards the camera, and that the set-up (camera height and distance) has to be fixed carefully.

2.4 Paper IV: Deep Learning and Kinematic Analysis of the Stroke-Affected Wrist

Oubre and Lee [4] combined deep learning with kinematic analysis to detect point-to-point movements of the stroke-affected wrist during functional activities and to assess them automatically. Their result shows that deep models can find meaningful movement segments in unstructured activity, and that kinematic measures computed from those segments can support automated rehabilitation assessment.

This paper motivates the combination used in the proposed system: kinematic features computed from the body keypoints (angles, range of motion, speed) together with a learned model that recognises the exercise and judges its quality. The kinematic features are easy to explain to a therapist, and the learned model adds the ability to deal with variations that fixed rules cannot handle.

2.5 Background Studies

Apart from these four papers, some basic references on the clinical side of stroke recovery, on pose estimation and on deep learning were also read. They are summarised here because the design of the system depends on them.

2.5.1 Clinical Assessment of Post-Stroke Motor Function

In the clinic, upper-limb recovery is assessed with standard scales. The Fugl-Meyer Assessment of the upper extremity [7] scores reflexes, voluntary movements inside and outside the movement synergies, wrist and hand function and coordination, and gives a motor score with a maximum of 66 points. The Action Research Arm Test scores grasp, grip, pinch and gross movements on a 57-point scale [8]. These scales are valid and widely accepted, but a trained therapist has to administer them, they take time, and some

items depend on the examiner's judgement. The proposed system does not try to replace these scales. It tries to provide a continuous, objective record of simple measures, such as the range of motion and the smoothness of the movement, between the clinic visits.

2.5.2 Human Pose Estimation

Human pose estimation (HPE) finds the position of body joints in an image. OpenPose [9] detects the keypoints of many people at once, using part affinity fields to join the points into skeletons. MediaPipe BlazePose [11] predicts 33 body landmarks and is designed to run in real time on a mobile processor. The YOLO family of detectors [13] has a pose variant that finds the person and regresses the 17 keypoints of the COCO format [12] in a single pass; YOLO11-Pose is evaluated for this purpose. For rehabilitation of the upper limb only the shoulders, elbows, wrists and hands (and the trunk for checking compensation) are needed, so any of these models gives enough points.

2.5.3 Deep Learning for Movement Sequences

A movement is a sequence of postures, so the models that suit it are those that handle sequences. Convolutional neural networks (CNN) [15] can learn short patterns along the time axis or over a map of joint coordinates. Recurrent networks, especially the Long Short-Term Memory (LSTM) [16] and the Gated Recurrent Unit (GRU) [17], keep a memory of previous frames and so can follow the whole repetition. Hybrid CNN–LSTM models use the first to extract local features and the second to follow their evolution. Generative Adversarial Networks (GAN) [18] can produce new, realistic trajectories and so can help when the amount of real stroke data is small.

2.5.4 Data Scarcity and Augmentation

All the papers agree on one difficulty: there are few labelled stroke movement data. Common ways to handle it are to use public benchmarks, to record a small own dataset with the help of a therapist, to augment the data by adding small noise, scaling and time-warping the keypoint sequences [21], and to use transfer learning. A combination of these will be used.

2.6 Summary of the Literature Survey

Table 2.1 lists the four papers side by side, with the methodology and what it means for the project.

2.7 Existing Systems and their Drawbacks

The methods that are used today for post-stroke rehabilitation can be grouped into three kinds.

- **Conventional therapist-based rehabilitation.** Exercises are demonstrated, supervised and assessed by a physiotherapist. The expert judgement is valuable, but the method needs continuous supervision and the assessment is subjective.

Table 2.1: Summary of the literature survey

Sl.	Paper details	Methodology / key findings	Relevance to this project
1	Han *et al.* (2025) [1]	Skeletal keypoints and optical flow, fused in a deep learning framework to separate post-stroke patients from healthy people	Supports keypoints as the main input; optical flow as an optional extra input
2	Robinson *et al.* (2025) [2]	Benchmark of stroke rehabilitation videos with 2D pose data for upper-limb assessment	Candidate dataset for training and validation
3	Wagh *et al.* (2025) [3]	MediaPipe used to track upper-limb reaching movements with a single camera	Confirms single-camera, low-cost pose tracking for the upper limb
4	Oubre and Lee (2024) [4]	Deep learning with kinematic analysis detected wrist movements of stroke patients	Supports combining kinematic features with a learned model

- **Wearable sensor-based systems.** Inertial sensors or electromyography (EMG) electrodes worn on the limb measure the movement directly. They give accurate data, but the patient has to wear and charge devices, the set-up takes effort, and the system depends on the sensors.
- **Robotic rehabilitation systems.** Powered devices guide or assist the affected limb and give precise, repeatable training, but they are expensive, bulky and found mainly in large centres, so home-based support is limited.

Table 2.2 compares these with the proposed camera-based approach.

Table 2.2: Comparison of rehabilitation approaches

Method	Advantage	Drawback
Therapist-based	Expert judgement and personal correction	Continuous supervision required; costly; subjective assessment
Wearable sensors	Direct measurement of motion	Patient must wear devices; sensor dependency; set-up effort
Robotic systems	Precise, repeatable assistance	High cost; bulky; limited home-based support
Proposed vision-based system	Needs only a camera; non-contact; suitable for home use	Depends on lighting, camera view and pose-estimation accuracy

2.8 Research Gap

From the survey three gaps are seen.

1. Much of the work on vision-based assessment classifies or measures movement offline and does not give corrective feedback to the patient while the exercise is being done.
2. Many systems depend on wearable sensors or robots, which limits their use at home and raises their cost.
3. Few works combine exercise recognition, quality scoring, real-time feedback and long-term progress tracking in one low-cost pipeline that a patient can run on a normal laptop.

A system is therefore proposed that uses a single camera and deep learning for recognition and assessment, that gives instant visual and audio guidance, and that stores the results so that progress can be seen over weeks.

2.9 Summary

The papers reviewed show that skeleton-based computer vision and deep learning can characterise post-stroke movement, that single-camera tracking with MediaPipe is practical for the upper limb, and that kinematic features combined with learned models can support automated assessment. The existing therapist-based, wearable and robotic approaches have the drawbacks shown in Table 2.2. The next chapter explains how the system is designed.

Chapter 3

SYSTEM DESIGN AND METHODOLOGY

This chapter explains how the proposed system works. The overall flow is first shown as a block diagram and the same system is then arranged in layers. The main blocks are then described one by one: the exercises and the capture set-up, pre-processing, pose estimation, feature extraction, the deep learning models and their training, the assessment and feedback logic, the progress-tracking database and the metrics used to judge the system. The hardware and software components that were selected are listed at the end of the chapter.

3.1 Block Diagram

Figure 3.1 shows the flowchart of the complete system. The patient performs a rehabilitation exercise in front of a camera. The system analyses the movement, decides whether it is correct, gives feedback and stores the result. The blocks are explained below.

1. **Patient performs rehabilitation exercise.** The patient carries out the exercise prescribed by the therapist, for example shoulder flexion, elbow flexion and extension or a reaching movement.
2. **Video capture.** A camera or webcam records the exercise.
3. **Video pre-processing.** Frames are extracted from the video, noise is removed and the images are enhanced.
4. **Pose estimation (skeleton detection).** Body keypoints are detected and the joints are tracked from frame to frame.
5. **Feature extraction.** Joint angles, range of motion, movement speed and other motion features are calculated.
6. **Deep learning model analysis.** CNN, LSTM or GRU models, hybrid models and feature learning are used to analyse the movement.
7. **Motor function assessment.** The output is converted into a performance score, an impairment indication and a recovery-level estimate.
8. **Decision: is the movement correct?** If yes, block 9A follows; if no, block 9B.

9A. **Good performance.** The patient is encouraged and the progress is updated.

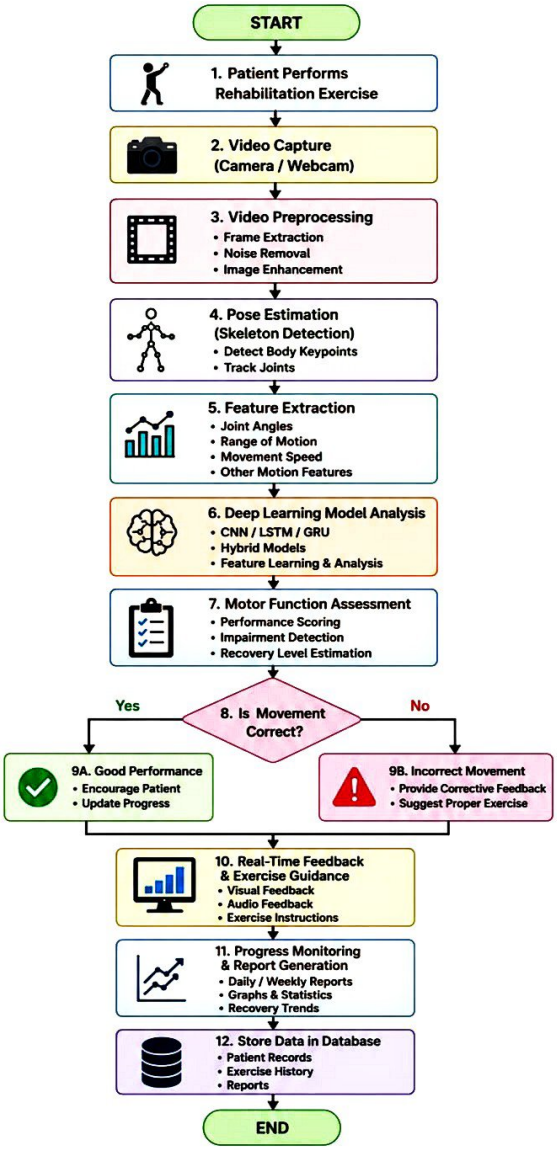


Figure 3.1: System block diagram / flowchart of the proposed system

9B. **Incorrect movement.** Corrective feedback is given and the proper exercise is suggested.

10. **Real-time feedback and exercise guidance.** Visual feedback, audio feedback and exercise instructions are presented.

11. **Progress monitoring and report generation.** Daily and weekly reports, graphs, statistics and recovery trends are produced.

12. **Store data in database.** Patient records, exercise history and reports are saved.

3.2 System Architecture

The block diagram describes the *order* in which things happen. Figure 3.2 shows the same system arranged in layers, to make clear which part runs where. The capture layer produces the frames, the processing layer turns them into keypoints, features and a score, the application layer shows the result to the patient and stores it, and the therapist can look at the stored data through a dashboard.

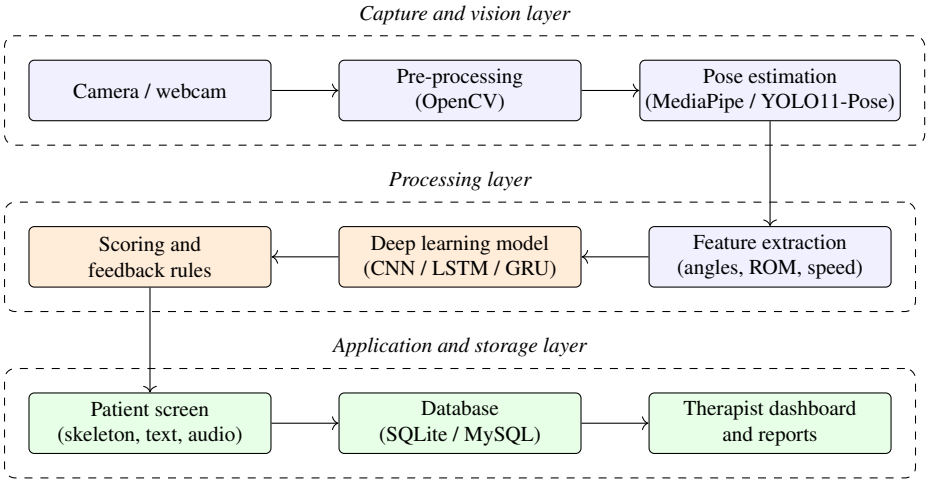


Figure 3.2: Layered architecture of the proposed system

All the processing for one patient session runs on one computer. The same program can be used in the clinic and at home. The only data that need to leave the computer are the session summaries, and these are sent only if tele-rehabilitation is enabled (see the future scope in Chapter 6).

3.3 Exercises and Data Capture

3.3.1 Exercise Set

The work starts with a small set of upper-limb exercises that are commonly given after a stroke and that can be seen clearly by a front-facing camera. Table 3.1 lists them with the three body points that define the angle to be measured, and the normal range of motion of a healthy adult from the usual orthopaedic tables. The normal values are given only for reference; the target range of a patient is set by the therapist and changes as the patient recovers.

Table 3.1: Initial exercise set and the angle used for each exercise

Exercise	Angle measured (joint at centre)	Normal range (reference)
Shoulder flexion	Hip – shoulder – elbow	0° to about 180°
Shoulder abduction	Hip – shoulder – elbow	0° to about 180°
Elbow flexion / extension	Shoulder – elbow – wrist	0° to about 150°
Wrist flexion / extension	Elbow – wrist – hand	about 80° / 70°
Forward reaching	Shoulder – elbow – wrist, with wrist path	Measured as path and reach

3.3.2 Recording Protocol

Each exercise is recorded for a fixed number of repetitions, at 25–30 frames per second. Before the first session, the patient (or the therapist) enters the affected side and the target range for each exercise. The camera placement for the recording is described in Section 4.3.

3.4 Video Pre-processing

A standard RGB camera at 25–30 frames per second is sufficient. Pre-processing is done with OpenCV [22] and has four steps: (i) frame extraction at a fixed rate; (ii) resizing of each frame to the input size expected by the pose model; (iii) noise reduction, for example with a Gaussian or a median filter; and (iv) illumination correction, for example by contrast limited histogram equalisation, so that the pose model works in dim rooms also.

3.5 Pose Estimation

3.5.1 Choice of Model

Two families of pose estimators are considered.

- **Landmark models** such as MediaPipe [10, 11] and OpenPose [9]. They return the 2D coordinates of body landmarks (and in MediaPipe a relative depth value as well) and run in real time on an ordinary laptop. MediaPipe is the first choice because it is light and free.
- **Detector-style pose models** such as YOLO11-Pose, which detect the person and predict the keypoints in a single forward pass [13]. They are very fast and easy to retrain on own data if the landmark model fails on patients with unusual postures.

3.5.2 Keypoint Processing

For every frame t, the output is a set of keypoints $\mathbf{p}_{j,t} = (x_{j,t}, y_{j,t})$ for joints $j = 1, \ldots, J$, each with a confidence value. The following steps prepare these points for analysis.

1. **Confidence check.** Points whose confidence is below a threshold are marked as missing.
2. **Interpolation.** Short gaps of missing values are filled by linear interpolation. Long gaps cause the repetition to be rejected, and the patient is asked to repeat it.
3. **Smoothing.** A Savitzky–Golay filter [20] or a moving average removes the small frame-to-frame jitter without flattening the real movement.
4. **Normalisation.** To make the data independent of the position of the patient and the distance from the camera, the points are expressed relative to the neck and scaled by the shoulder width:

$$\tilde{\mathbf{p}}_{j,t} = \frac{\mathbf{p}_{j,t} - \mathbf{p}_{\text{neck},t}}{\left\| \mathbf{p}_{\text{shoulderL},t} - \mathbf{p}_{\text{shoulderR},t} \right\|} \tag{3.1}$$

3.6 Feature Extraction and Design Calculations

The following kinematic features are calculated from the normalised keypoints.

3.6.1 Joint Angle

For a joint B with a proximal neighbour A and a distal neighbour C (for example shoulder, elbow and wrist for the elbow, as in Figure 3.3), the angle at B is

$$\theta_B = \cos^{-1}\left(\frac{(\mathbf{A}-\mathbf{B})\cdot(\mathbf{C}-\mathbf{B})}{\|\mathbf{A}-\mathbf{B}\|\,\|\mathbf{C}-\mathbf{B}\|} \right) \tag{3.2}$$

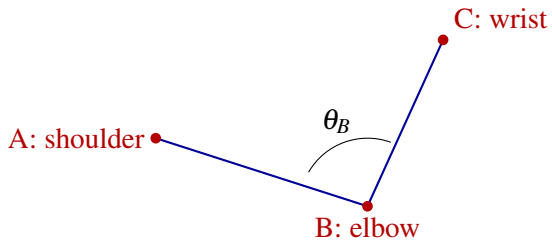


Figure 3.3: Angle θ_B at the elbow computed from three keypoints

3.6.2 Range of Motion and Speed

The range of motion (ROM) over one repetition is the difference between the largest and the smallest angle:

$$\text{ROM} = \theta_{\max} - \theta_{\min} \tag{3.3}$$

The movement speed of a joint (for example the wrist) at frame t with frame interval $\Delta t = 1/\text{FPS}$ is

$$v_t = \frac{\|\tilde{\mathbf{p}}_t - \tilde{\mathbf{p}}_{t-1}\|}{\Delta t} \tag{3.4}$$

3.6.3 Smoothness, Symmetry and Repetitions

A healthy movement is smooth, while a movement after a stroke is often jerky. As a simple measure of smoothness the mean squared jerk (the third derivative of position) is used, normalised by the duration and the amplitude of the movement; a smaller value means a smoother movement. The symmetry between the affected and the unaffected arm is measured as the ratio of their ROM values. Repetitions are counted by finding the peaks of the joint-angle signal after smoothing, with a minimum peak height and a minimum distance between peaks, so that small tremors are not counted. The sequence of features over a window of T frames ($T = 60$, about two seconds) forms the input of the learning model.

3.7 Deep Learning Models

Five kinds of models are compared for exercise recognition and quality assessment. All of them take as input the window of T frames described above, in which each frame has the normalised coordinates of the upper-limb keypoints and the joint angles.

3.7.1 Convolutional Neural Network (CNN)

A CNN [15] learns local patterns by sliding small filters over its input. If the keypoint sequence is arranged as a map of time $\times$ joint coordinates, a convolution layer computes

$$y_{i,j} = \sigma\left(\sum_m \sum_n w_{m,n}\, x_{i+m,\, j+n} + b\right) \tag{3.5}$$

where w are the filter weights, b is the bias and σ is a non-linear activation such as ReLU. Pooling layers reduce the size, and fully connected layers with a softmax output give the class probabilities. A CNN is fast to train and to run, and it suits the spatial pattern of the pose.

3.7.2 LSTM and GRU

Rehabilitation movements are sequences, so recurrent networks are natural. The LSTM [16] keeps a cell state c_t that is controlled by forget, input and output gates:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{3.6}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{3.7}$$
$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{3.8}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{3.9}$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{3.10}$$
$$h_t = o_t \odot \tanh(c_t) \tag{3.11}$$

The GRU [17] is a lighter variant that has only an update gate and a reset gate. Both models follow the development of the joint angles during a repetition, and they can also be trained to predict the next position of a joint, such as the wrist trajectory shown in the preliminary plots of Chapter 5.

3.7.3 Hybrid CNN–LSTM Model

The hybrid model first uses convolution layers to extract local features from each short part of the sequence and then passes the feature sequence to an LSTM or GRU layer that models how the features change in time. It combines the advantages of both and is the model expected to give the best balance between accuracy and speed.

3.7.4 YOLO-based Pose Model

YOLO11-Pose detects the person and predicts the upper-limb keypoints directly from the image in one pass, giving real-time speed. Its quality is measured by precision, recall and mean average precision (mAP). If the landmark model is not accurate enough for patients with unusual postures, YOLO11-Pose can be fine-tuned on own annotated frames.

3.7.5 Generative Adversarial Network (GAN)

A GAN [18] contains a generator G and a discriminator D that are trained against each other:

$$\min_G \max_D \; \mathbb{E}_x[\log D(x)] + \mathbb{E}_z[\log(1 - D(G(z)))] \tag{3.12}$$

In this project a GAN is considered for generating realistic upper-limb trajectories. They can be used to enlarge the small stroke dataset and as a reference trajectory of a normal movement.

3.7.6 Initial Model Configurations

Table 3.2 gives the starting configuration of the three classification models. These are first choices that will be changed after the experiments of the eighth semester.

Table 3.2: Initial configuration of the classification models

Model	Layers (input: $T \times F$ window, $T = 60$)	Output
1-D CNN	Conv1D(64, k=5) → BatchNorm → ReLU → MaxPool; Conv1D(128, k=3) → ReLU; global average pooling; Dense(64); dropout 0.3	Softmax over K classes
LSTM	LSTM(64, return sequences) → LSTM(64); dropout 0.3; Dense(32)	Softmax over K classes
Hybrid	Conv1D(64, k=5) → ReLU → MaxPool; LSTM(64); Dense(32); dropout 0.3	Softmax over K classes

3.8 Training Strategy

- **Data split.** The data are divided into training, validation and test sets in the ratio of about 70:15:15. The split is made *by subject*, so that the frames of one person are never in both the training and the test set; otherwise the result would look better than it really is.

- **Augmentation.** Small random noise and scaling are added to the keypoints, the sequences are stretched or compressed slightly in time, and left-right mirroring is used to simulate the opposite affected side [21].
- **Loss and optimiser.** For classification the cross-entropy loss is used:

$$L = -\sum_{k=1}^{K} y_k \log \hat{y}_k \tag{3.13}$$

  where y_k is the true label and $\hat{y}_k$ the predicted probability of class k; for trajectory prediction the mean absolute error is used. The Adam optimiser [19] is used with an initial learning rate of 10^{-3} and a batch size of 32.
- **Regularisation.** Dropout, early stopping on the validation loss and class weights (if the classes are unequal) are used to limit over-fitting.

3.9 Motor Function Assessment and Feedback

3.9.1 Scoring

The assessment module converts the model output and the kinematic features into a score between 0 and 100. A weighted combination is proposed:

$$S = 100 \sum_{k=1}^{K} w_k s_k, \qquad \sum_{k=1}^{K} w_k = 1, \quad 0 \le s_k \le 1 \tag{3.14}$$

The sub-scores s_k are listed in Table 3.3. The weights w_k will be tuned with the guidance of the therapist, and the movement is declared correct when S exceeds a threshold that the therapist can set for each patient.

Table 3.3: Sub-scores of the performance score (weights are initial values)

Sub-score	Definition	Initial weight
s_1 Range of motion	$\min(\mathrm{ROM}/\mathrm{ROM}_{\mathrm{target}}, 1)$	0.35
s_2 Smoothness	$1 - \min(J/J_{\mathrm{ref}}, 1)$, J = normalised jerk	0.20
s_3 Speed	1 inside the prescribed speed band, falling towards 0 outside	0.15
s_4 Symmetry / trunk	penalty for trunk lean or for a large difference from the healthy side	0.10
s_5 Model confidence	probability of the class “correct” from the neural network	0.20

3.9.2 Feedback Rules

When the score is below the threshold, the sub-score that is lowest decides the message that is shown, so that the patient receives one clear instruction at a time. Examples are given in Table 3.4.

Table 3.4: Examples of corrective feedback rules

Condition detected	Message to the patient
ROM below target	“Try to lift your arm a little higher.”
Movement too fast	“Slow down and move smoothly.”
Trunk leaning during the exercise	“Keep your back straight against the chair.”
Shaky or jerky movement	“Move steadily; rest if you feel tired.”
Score above threshold	“Well done. Repeat the movement.”

The feedback is shown as a skeleton drawn over the live video, in which the joints are coloured green when the movement is correct and red when it is not, together with a short text and an audio prompt. If the system sees pain behaviour or an unstable posture, it will stop the exercise and ask the patient to rest. The messages will be reviewed by the therapist before being used with patients.

3.10 Progress Tracking and Database

At the end of every exercise, the system stores a summary of the session. The main tables are *patient*, *exercise*, *session* and *report*, which are described in Chapter 4. From the stored values the module draws graphs of the ROM, the score and the number of repetitions over days and weeks, and it produces a daily and a weekly report for the patient and a more detailed one for the therapist. A rising trend of the ROM and the score is the objective sign of recovery that the therapist wants to see.

3.11 Performance Metrics

The models and the complete system are evaluated using the following measures:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, \quad \text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN} \tag{3.15}$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N} |p_i - \hat{p}_i|, \qquad \text{mAP} = \frac{1}{C}\sum_{c=1}^{C} \text{AP}_c \tag{3.16}$$

Here TP, TN, FP, FN are the true and false positives and negatives, p_i and $\hat{p}_i$ are the true and predicted (normalised) keypoint coordinates and AP_c is the average precision of class c. In addition, the error between the angle given by the system and a reference angle (goniometer or hand-marked video frames) and the delay between the movement and the feedback on the screen will be measured.

3.12 Component Selection

The complete hardware consists of a camera, a computer and a display. Nothing needs to be worn by the patient. Table 3.5 lists the hardware components and the purpose of

each. Training of the deep learning models needs more computing power than their use, so the training is planned on a computer with a GPU (a laboratory computer or a free cloud notebook), and the trained model is run on a normal laptop. The trained model is saved as a file and copied to the laptop.

Table 3.5: Hardware components

Component	Purpose and requirement
Camera / webcam	Captures the patient's movements. A 720p (1280×720) camera at 30 frames per second is sufficient; a fixed focus with automatic exposure is preferred
Computer / laptop	Runs the pose estimation, the deep learning inference and the user interface. A modern multi-core processor with 8 GB of memory is the minimum recommended
GPU (optional)	Speeds up the training of the models and the real-time inference. Not needed for running the trained model on a laptop
Storage device	Stores the datasets, the trained models and the patient records
Monitor	Displays the live video, the skeleton overlay and the feedback; a larger screen is easier for elderly patients
Internet connection	Needed for installing the libraries and models, and for tele-rehabilitation in the future

Table 3.6 lists the software tools. All of them are free and widely used, which keeps the cost of the project low and makes the work easy to reproduce. The technologies used are computer vision, deep learning, machine learning and pose estimation.

3.13 Summary

The pipeline converts camera video into keypoints, kinematic features, a learned assessment, feedback and stored progress records, and the hardware and software needed to run it on a normal laptop were selected. The next chapter describes how the design is realised: the software modules, the session flow, the database, the screens and the set-up.

Table 3.6: Software components

Tool	Use in the project
Python	Main programming language for all the modules
OpenCV	Video capture, frame extraction, filtering and drawing of the skeleton overlay
MediaPipe / OpenPose	Pose estimation and keypoint extraction
Ultralytics YOLO11-Pose	Alternative detector-style pose model, which can be fine-tuned
TensorFlow (Keras) / PyTorch	Building, training and running the deep learning models
NumPy, Pandas, SciPy	Numerical computation, feature tables, filtering and peak detection
scikit-learn	Metrics (accuracy, precision, recall, confusion matrix) and data splitting
SQLite / MySQL	Database for patient records, exercise history and reports
Flask / Streamlit	Web-based user interface for the patient and the therapist
Matplotlib	Graphs of progress and the plots of the results

Chapter 4

IMPLEMENTATION

The system is mostly software, but it still needs a small hardware set-up and a well-prepared software environment. This chapter describes how the design of Chapter 3 is realised: the software modules, the flow of one exercise session, the database, the user screens, the tools used and the practical points of the set-up. The results are given in Chapter 5.

4.1 Software Setup

4.1.1 Modules

The program is divided into small modules with clear inputs and outputs, so that every member of the team can develop and test one module independently. Table 4.1 lists them.

Table 4.1: Software modules of the system

Module	Input	Output
Capture	Camera stream	Frames with time stamps
Pre-processing	Frames	Cleaned, resized frames
Pose estimation	Cleaned frames	Keypoints with confidence values
Feature extraction	Keypoint sequence	Angles, ROM, speed, smoothness, repetition count
Model inference	Window of features	Exercise class and quality probability
Assessment	Features and model output	Score and feedback message
Feedback display	Score, message, frame	Live screen with skeleton overlay and audio
Progress and report	Session summaries	Graphs and daily/weekly reports

4.1.2 Flow of One Exercise Session

Figure 4.1 shows what happens in one session. The patient first selects the exercise. The system then repeats a loop for every captured frame until the required number of repetitions has been completed, and finally it stores the summary.

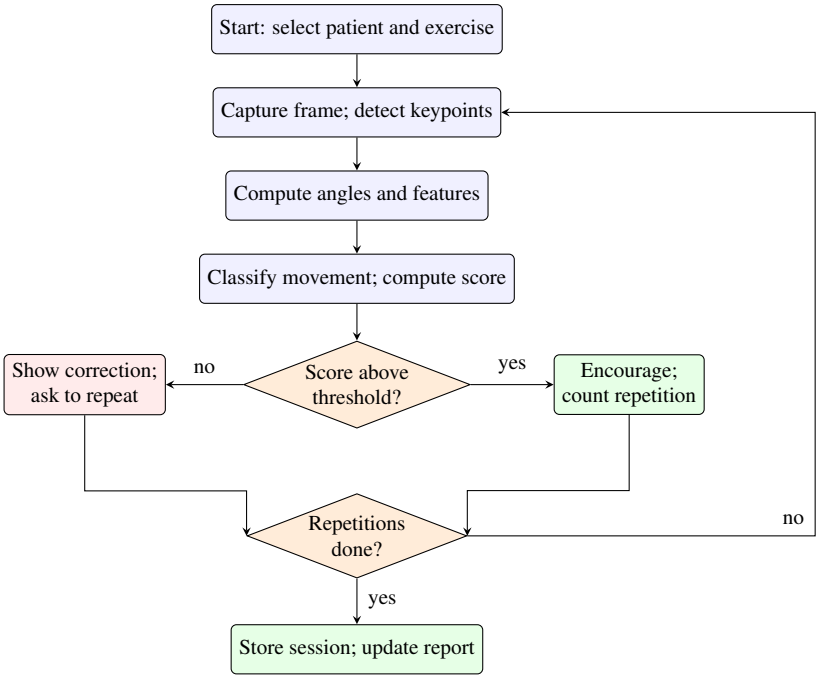


Figure 4.1: Flow of one exercise session

4.1.3 Database Design

The database keeps only what is needed for the progress report. The patient is identified by a code and not by a name where possible. Table 4.2 gives the main tables and their fields; the SQL schema is given in Appendix D.

Table 4.2: Main tables of the database

Table	Main fields
patient	patient_id, code/name, age, affected_side, therapist_id, start_date
exercise	exercise_id, name, joint_triplet, target_rom, target_repetitions, speed_band
session	session_id, patient_id, exercise_id, date_time, repetitions, mean_score, max_rom, mean_speed, smoothness, duration
feedback_log	log_id, session_id, time_stamp, message, sub-score that triggered it
report	report_id, patient_id, period (daily/weekly), summary values, created_on

4.1.4 User Screens

The patient screen must be simple: a large live picture with the skeleton, a single line of text and a count of repetitions, and large buttons. The therapist screen is a dashboard that lists the patients, shows the progress graphs and allows the exercises and targets to be changed. Figure 4.2 shows the planned layout of the two screens.

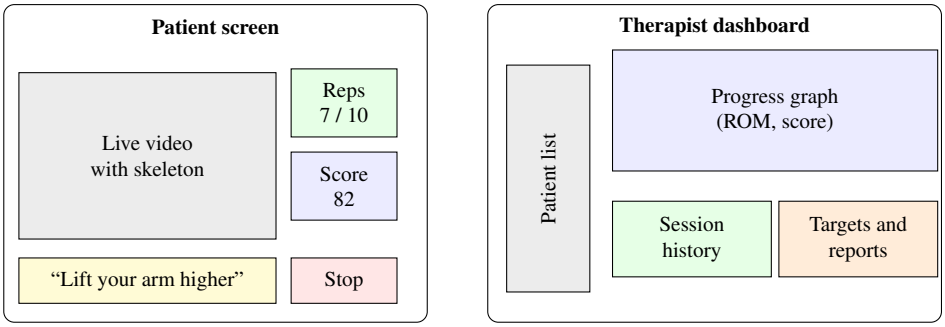


Figure 4.2: Planned layout of the patient screen and the therapist dashboard

4.2 Components and Tools Used

4.2.1 Camera and Computer

A 720p webcam at 30 frames per second is used with a laptop having a multi-core processor and 8 GB of memory (Table 3.5). The patient wears nothing. A GPU is used only for training the models.

4.2.2 Software Tools

All the software tools of Table 3.6 are free. Python is the language of all the modules; OpenCV is used for capture, filtering and the skeleton overlay; MediaPipe is the main pose estimator, with YOLO11-Pose for comparison; TensorFlow/PyTorch is used for the models; SQLite stores the records; and Flask or Streamlit gives the web-based screens.

4.2.3 Reference Code

Reference sketches of the main modules are given in the Appendix. The feature-extraction and scoring code (Appendix B) and the database schema (Appendix D) have been run on test data. The pose-extraction code (Appendix A) and the LSTM and hybrid models (Appendix C) show the structure of the modules and are to be completed and tested in Semester VIII.

4.3 Complete Setup

The camera is placed in front of the patient at about shoulder height, at a distance of about 1.5 to 2.5 m so that both arms and the upper trunk are in the picture. The lighting should be even and the background plain where possible. The patient sits on a chair without arm rests or stands, wearing clothes that do not hide the arms. The practical points of the set-up are:

- **Lighting and background.** Strong backlight (a window behind the patient) makes the pose estimate unreliable. The patient should face the light, and a plain background helps.
- **Clothing and occlusion.** Loose sleeves or a shawl that hides the elbow reduce the accuracy. The system will warn when the confidence of a keypoint remains low.

- **Camera stability.** The camera is fixed on a stand or on the monitor. Moving the camera during an exercise changes the measured angles.
- **Response time.** The pose model, the feature calculation and the classifier together must run at the frame rate of the camera on the target laptop. If the speed is not enough, the resolution of the input will be reduced or a lighter model used.
- **Privacy and safety.** The video is processed on the same computer and is not stored by default. The first sessions of every patient should be done with the therapist or a family member present.

4.4 Summary

The system is built from small modules with defined inputs and outputs, a session loop, a small database and two simple screens, using free tools on an ordinary laptop and webcam. The practical conditions of the set-up were listed. The next chapter presents the work completed in Semester VII and the preliminary results.

Chapter 5

RESULTS AND DISCUSSION

This chapter presents the work completed in Semester VII and discusses the preliminary results. The work done so far is mainly the study, the design and the selection of tools; the implementation and the training of the models continue in Semester VIII. The preliminary comparison plots in this chapter are *illustrative* and are not measured results.

5.1 Work Completed in Semester VII

5.1.1 Activities Completed

The activities are listed in Table 5.1. The status column shows whether an activity has been completed or is still going on.

The plan for July 2026, which covers the first four activities, is shown in Table 5.2, and the overall timeline of the project is shown in Figure 5.1.

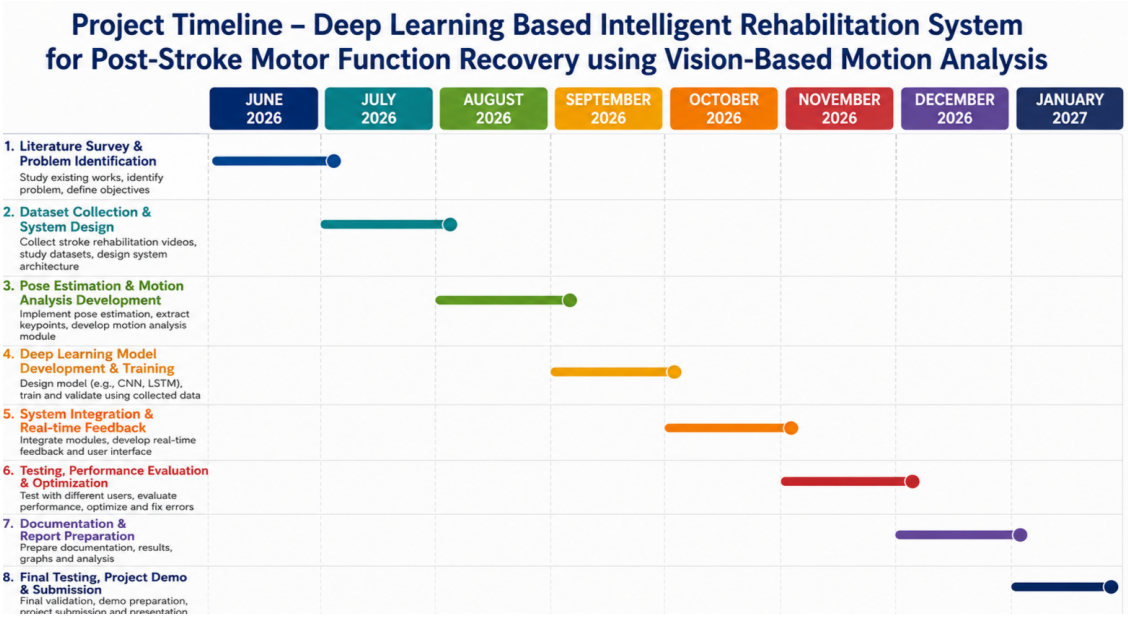


Figure 5.1: Project timeline

Table 5.1: Project status in the seventh semester

Sl.	Activity	Outcome	Status
1	Team formation and guide allocation; topic selection (June 2026)	Project title finalised; problem identified; planning and timeline prepared	Completed
2	Literature survey (June–July 2026)	Four recent papers and background references studied; research gap identified	Completed
3	Problem statement and objectives (end of June 2026)	Problem statement and eight objectives finalised; proposal presented	Completed
4	Methodology and system design (July 2026)	Block diagram and flowchart prepared (Figure 3.1); initial architecture defined	Completed
5	Selection of hardware and software (July 2026)	Camera, laptop, Python, OpenCV, MediaPipe, TensorFlow/PyTorch, database and interface tools selected	Completed
6	Study of candidate algorithms	CNN, LSTM, YOLO11-Pose and GAN compared with preliminary plots (Section 5.2)	Completed
7	Dataset collection and pre-processing	Public benchmark and own recordings being collected	Ongoing
8	Pose estimation and motion-analysis module	Keypoint extraction and feature calculation being implemented	Ongoing

Table 5.2: Project plan for July 2026

Week	Activity
Week 1	Literature survey and review of research papers related to deep neural networks
Week 2	Problem identification, research gap analysis and definition of project objectives
Week 3	Software set-up and preparation of the initial system architecture
Week 4	Selection of hardware and software components

5.1.2 Achievements against the Objectives

1. **A clear problem and scope.** The need for automated, home-based supervision of stroke rehabilitation was established from the literature, and the scope (upper limb, single camera) was fixed.
2. **A complete design.** A twelve-block flowchart, a layered architecture, the module list, the database tables and the screens were designed, so that the implementation can start in parallel on several modules.
3. **Tools selected.** The hardware and software were chosen so that the system can run on a normal laptop with free software.
4. **Algorithm options compared.** Four algorithms were compared at the level of expected behaviour, which tells where to concentrate the experiments.
5. **Evaluation plan.** The metrics, the data split and the scoring rules were defined in advance, so that the results of the eighth semester will be comparable.

5.1.3 Test of the Reference Code

The feature-extraction and scoring sketch (Appendix B) was run on a synthetic angle signal $\theta(t) = 90^\circ + 40^\circ \sin(2\pi t/2)$, sampled at 30 frames per second for 6 seconds, which represents three repetitions with a range of motion of 80°. Table 5.3 shows that the code returned the expected values. The sub-scores used for the score test are sample values and not measurements of a patient.

Table 5.3: Output of the reference code on test data

Test	Expected	Output
Angle between $(0,1)$, $(0,0)$, $(1,0)$	90°	90.0
Repetitions in the synthetic signal	3	3
Range of motion after smoothing	80°	80.0
Score for ROM $= 70$, $\text{ROM}_{\text{target}} = 80$ and sample sub-scores 0.8, 1.0, 0.9, 0.85	87.6 by Equation (3.14)	87.6

5.2 Preliminary Analysis of Candidate Algorithms

Four candidate algorithms (LSTM, CNN, GAN and YOLO11-Pose) were studied for the upper-limb rehabilitation task. For each of them, four panels are plotted: (A) training and validation loss, (B) accuracy or detection metrics, (C) predicted versus actual wrist trajectory and (D) per-joint mean absolute error (MAE) over the upper-limb joints.

Note: these plots were prepared for the third review presentation to illustrate the expected behaviour of each algorithm and the layout of the results. The values in them are illustrative and will be replaced by the measured results when the models are trained on the final dataset in the eighth semester.

Table 5.4 gathers the end-of-training values that can be read from the illustrative plots, so that the four algorithms can be compared easily.

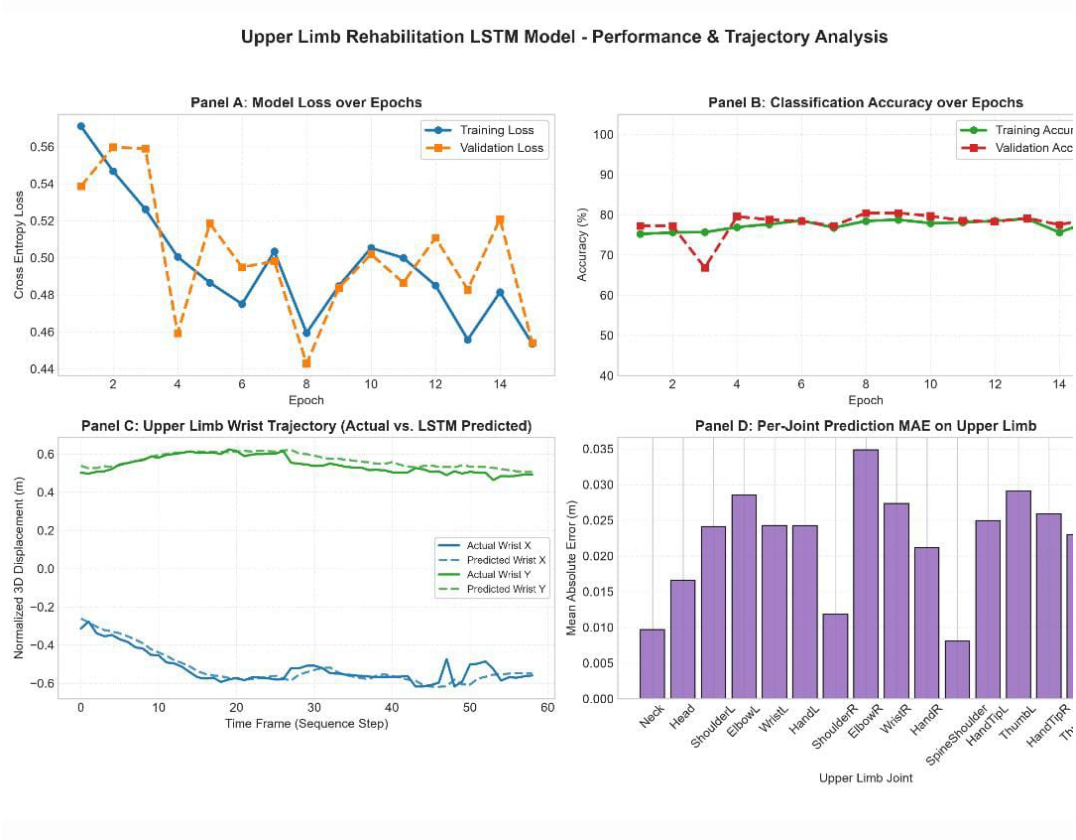


Figure 5.2: LSTM model: loss, accuracy, wrist trajectory and per-joint MAE (illustrative)

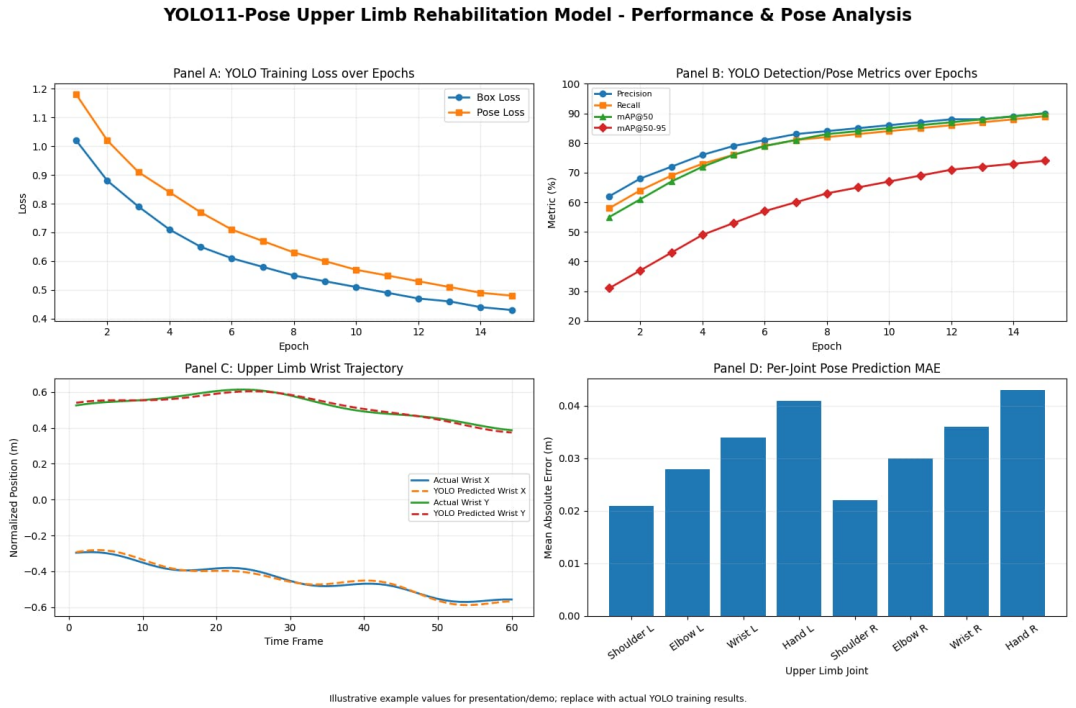


Figure 5.3: YOLO11-Pose model: training loss, detection metrics, wrist trajectory and per-joint MAE (illustrative)

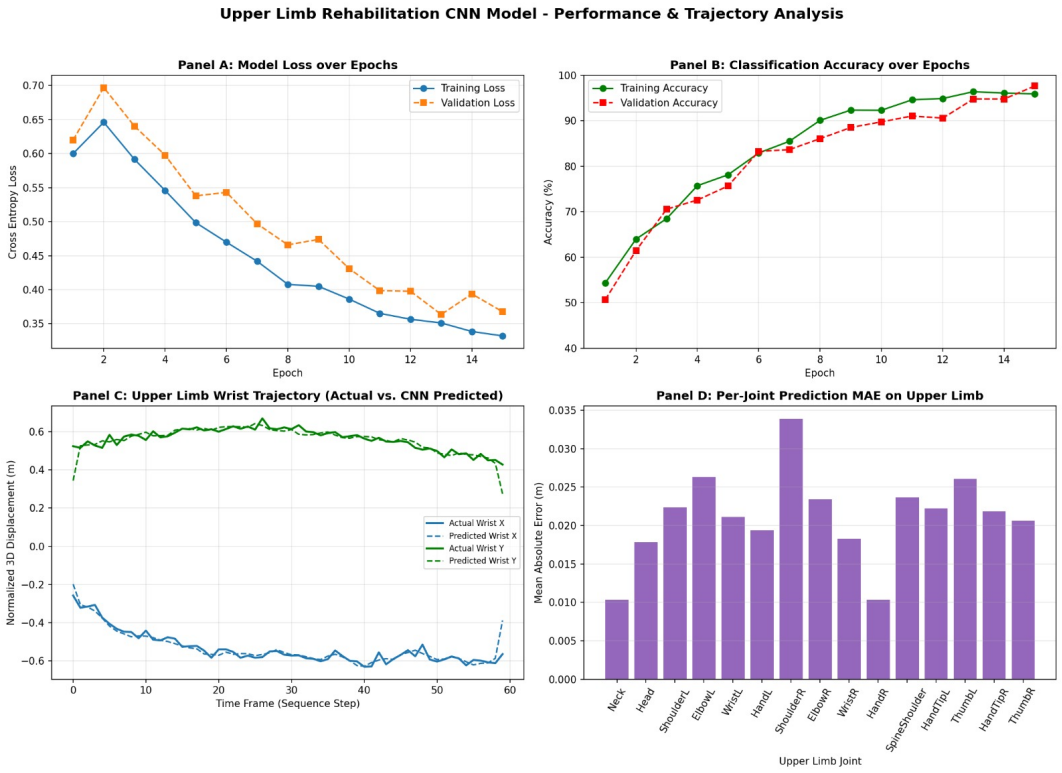


Figure 5.4: CNN model: loss, accuracy, wrist trajectory and per-joint MAE (illustrative)

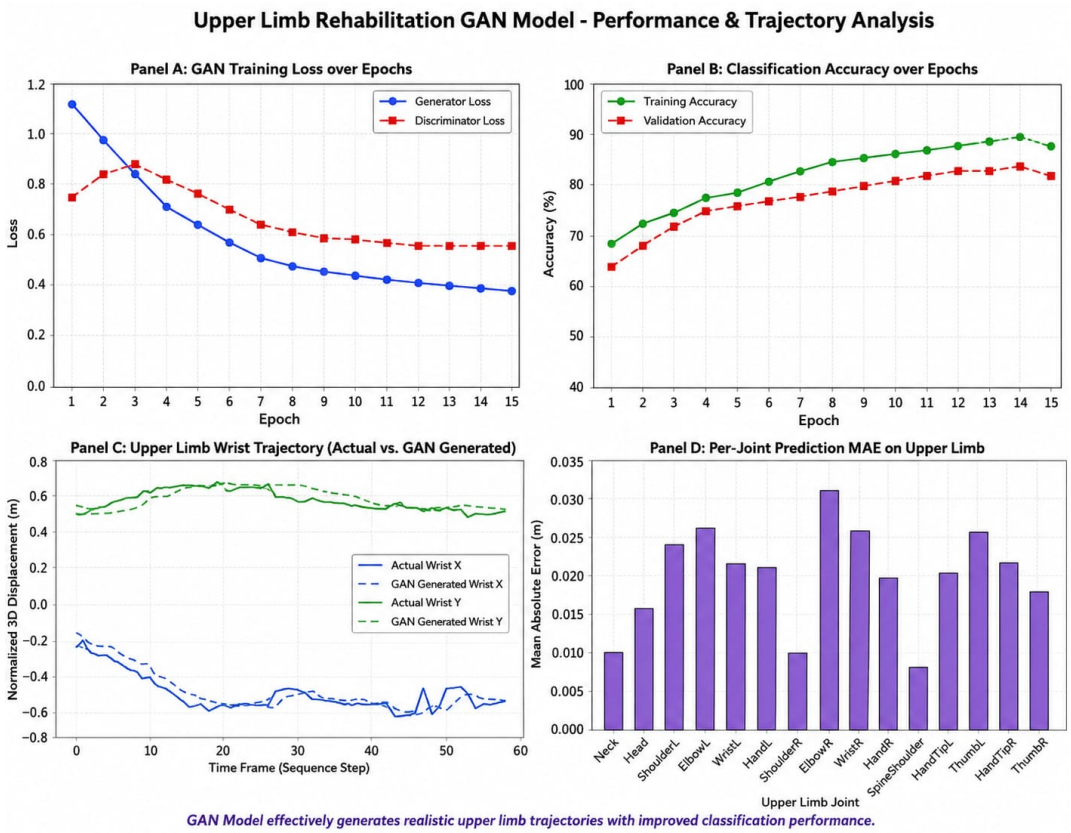


Figure 5.5: GAN model: training loss, accuracy, wrist trajectory and per-joint MAE (illustrative)

Table 5.4: Comparison of the candidate algorithms (values read from the illustrative plots)

Algorithm	Approximate value at the end of training	Remark
LSTM	Accuracy about 77–80%	Loss fluctuates; follows the wrist path well
YOLO11-Pose	Precision and mAP@50 about 90%; mAP@50–95 about 74%	Smooth losses; suited to real-time keypoints
CNN	Training about 96%, validation about 97%	Highest accuracy; close trajectory match
GAN	Training about 88%, validation about 82%	Realistic trajectories; gap between training and validation

5.3 Discussion of the Results

In the illustrative plots, the CNN reaches the highest classification accuracy (above 90% at the end of training), with falling loss and a close match between the predicted and the actual wrist trajectory (Figure 5.4). The LSTM follows the wrist trajectory well, but its loss fluctuates and its accuracy remains lower (Figure 5.2). The YOLO11-Pose plots show smoothly decreasing losses with precision and mAP@50 near 90% and a lower mAP@50–95 (Figure 5.3), which suggests that YOLO is attractive for real-time keypoint detection. The GAN generates realistic trajectories, with some gap between the training and the validation accuracy (Figure 5.5). In all four cases the error is largest at the distal joints (wrist, hand and fingers) and smallest at the neck and shoulders. This is as expected, because the distal joints move more and are harder to detect.

On this basis a combination is planned: a pose model (MediaPipe or YOLO11-Pose) for real-time keypoints, followed by a CNN or hybrid CNN–LSTM classifier for the quality of the exercise, with the GAN used for data augmentation. The final choice will be made from the measured results and not from the illustrative plots.

Three things noticed during this semester will influence the next stage. First, good data are the main problem, so a public benchmark will be combined with a small set of own recordings. Second, the camera position and the lighting matter as much as the choice of the model. Third, the plots of the candidate algorithms are only a guide; the real comparison can be made only when all models are trained on the same data with the same split. A limitation of the work so far is that no measurement on a real patient or volunteer has been made.

5.4 Summary

The study, the design and the tool selection are complete, and the reference code for features and scoring gave the expected output on test data. The data collection and the pose-estimation module are in progress. The comparison of the candidate algorithms is only illustrative. The next chapter gives the conclusion and the plan for Semester VIII.

Chapter 6

CONCLUSION AND FUTURE WORK

This chapter summarises the work done in Semester VII, gives the plan for Semester VIII with the risks and the expected results, and suggests how the work can be extended. The mapping of the project with the POs, PSOs and SDGs follows at the end of the chapter.

6.1 Conclusion

The project proposes a camera-based, deep learning rehabilitation system that provides automated assessment of post-stroke motor recovery, accurate motion analysis through computer vision and deep learning, real-time feedback and exercise guidance, and improved effectiveness of post-stroke rehabilitation without wearable sensors or robots. In the seventh semester the problem was defined, the literature was surveyed, the system was designed and the tools were selected (objectives 1 to 4 of Chapter 1 are addressed at the level of design). The reference code for features, repetition counting and scoring gave the expected output on test data. The implementation, training and validation of the system, together with its testing with different users, will be completed in the eighth semester, so that the final system can be demonstrated and submitted by March 2027.

6.2 Work Plan for Semester VIII

Table 6.1 gives the month-wise plan for the remaining part of the project, and Figure 6.1 shows it as a chart.

6.2.1 Detailed Work Plan

Dataset and pose estimation. The public benchmark [2] will be studied and, if its licence permits, used for training. A small set of own recordings will be made, with the permission of the participants and the guidance of the therapist, in conditions like those of a home. The frames will be passed through MediaPipe (and YOLO11-Pose for comparison) and the keypoints will be stored. The accuracy of the keypoints will be checked against hand-marked frames.

Model development. The CNN, the LSTM/GRU and the hybrid model will be trained with the same split and compared by accuracy, precision, recall and the confusion matrix. The GAN will be tried for data augmentation. The best model will be selected after taking

Table 6.1: Project plan for the eighth semester

Sl.	Period	Planned activity
1	November 2026	Complete the pose-estimation and feature modules; train the first models; prepare the research paper based on the project work; plagiarism check
2	December 2026	System implementation and integration; real-time feedback screen; advanced rehabilitation assessment module (scoring and reports)
3	January 2027	Final implementation and system validation; comparison with existing rehabilitation methods; documentation of the final report
4	February–March 2027	Testing with different users, optimisation and fixing of errors; final project demonstration with live exercise monitoring; final performance evaluation and validation; final report and project submission

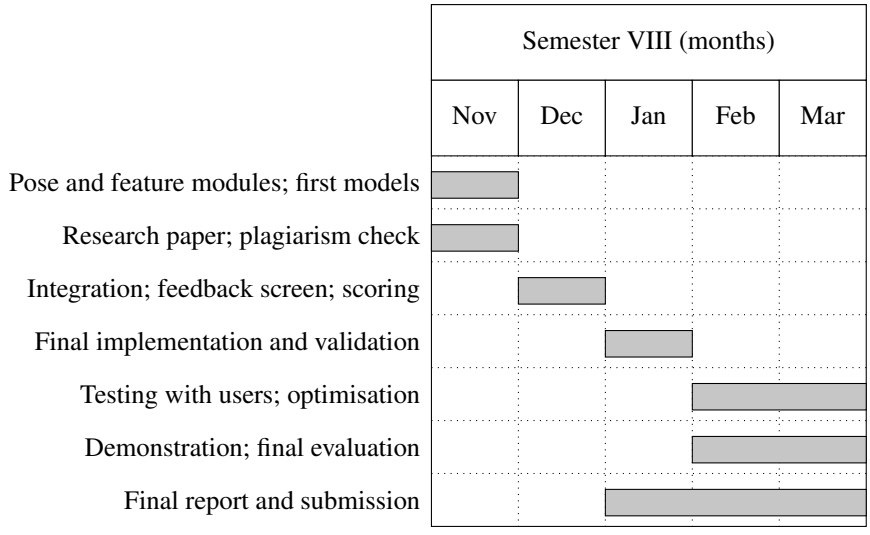


Figure 6.1: Work plan for Semester VIII

the speed also into account. The illustrative plots of this report will be replaced by the real ones.

System integration. The modules will be joined in the program described in Chapter 4, with the live skeleton overlay, the messages, the audio prompts and the repetition counter. The database and the report generator will be connected so that every session is stored.

Performance evaluation. The following will be measured:

- **Angle accuracy:** the difference between the angles measured by the system and the reference angles.
- **Classification performance:** accuracy, precision and recall of the exercise and quality classifier on the test subjects.
- **Latency:** the time from the movement to the message on the screen, and the frame rate on the laptop.
- **Usability:** a short trial with volunteers (and, if permitted, with a therapist) to see whether the screen is easy to follow.
- **Agreement with the therapist:** the agreement between the score of the system and the judgement of the therapist for the same movements.

Improvements and documentation. Depending on the results and the time, the following improvements will be considered: optical flow as a second input [1, 14], a voice prompt in the local language, a larger exercise set including wrist and finger exercises, and a simple mobile version of the screen. A research paper on the system will be prepared, and the final report will include the complete results, the photographs of the set-up, the source code and the user manual.

6.2.2 Risks and Mitigation

Table 6.2 lists the main risks that can affect the completion of the work, with the planned actions.

6.2.3 Expected Results

- Accurate vision-based motion tracking.
- Automatic motor function assessment.
- Detection of abnormal movement patterns.
- Personalised rehabilitation recommendations.
- Reduced therapist workload.
- Continuous progress monitoring and report generation.

Table 6.2: Risks and mitigation

Risk	Effect	Mitigation
Too little or unbalanced data	Over-fitting; poor accuracy on new users	Public benchmark plus own recordings; augmentation; subject-wise split
Poor lighting, occlusion or camera view	Unreliable keypoints	Fixed set-up guide; confidence check; warning to the user
Real-time speed on a laptop	Delayed feedback	Lighter model; lower input resolution; skip frames
Differences between healthy and patient movement	Model fails on real patients	Include patient data; therapist review of labels and feedback
Safety and consent	Harm to patient; ethical problems	Therapist-set limits; consent and approval before any patient trial
Schedule delays	Incomplete evaluation	Parallel work on modules; weekly review with the guide

6.3 Future Scope

Once the basic system works, it can be extended in several directions:

- **AI-driven personalised rehabilitation programmes** that change the exercises and the targets automatically according to the progress of each patient.
- **VR/AR-assisted rehabilitation,** where the exercises are presented as games to make them more motivating.
- **Integration with wearable sensors and EMG** to add information on muscle activity to the camera data.
- **Cloud-based tele-rehabilitation,** where the reports are sent to the therapist at a distance and the programme can be adjusted without a visit.

6.4 Mapping with POs and PSOs

The project is mapped with the Programme Outcomes (POs) and the Programme Specific Outcomes (PSOs) of the department in Table 6.3.

Mapping levels: **1** – Low, **2** – Medium, **3** – High.

Table 6.3: Mapping of the project with POs and PSOs

PO / PSO	**Level**	**Justification**
PO1: Engineering Knowledge	3	Mathematics, image and signal processing and machine learning are used for keypoints, joint angles and the sequence models.
PO2: Problem Analysis	3	The gaps in home-based rehabilitation were identified from the literature and stated as the problem and objectives.
PO3: Design/Development of Solutions	3	A pipeline, scoring rules, database and screens were designed with patient safety and privacy in mind.
PO4: Conduct Investigations of Complex Problems	2	A subject-wise data split and metrics were defined for comparing the models; the experiments follow in Semester VIII.
PO5: Engineering Tool Usage	3	OpenCV, MediaPipe, YOLO11-Pose, TensorFlow/PyTorch and SQLite were selected and used.
PO6: The Engineer and the World	3	The system supports stroke survivors at home, with lower cost and travel and with the video kept on the patient's computer.
PO7: Ethics	2	Patient privacy, consent, therapist approval and dataset licences are part of the design.
PO8: Individual and Collaborative Team Work	2	The work is divided into modules that each member develops and tests independently.
PO9: Communication	2	Review presentations, this report and a planned research paper.
PO10: Project Management and Finance	2	A month-wise plan, a risk table and a low-cost, free-software design were prepared.
PO11: Life-Long Learning	3	New tools were learnt: pose estimation, sequence models and a GAN for data augmentation.
PSO1: Analysis and design of electronic circuits, VLSI systems and embedded architectures	1	The hardware is only a camera and a laptop; embedded deployment is a possible extension.

PO / PSO	Level	Justification
PSO2: Solving electronics and communication problems with hardware, software and modern tools	3	A camera, computer and software tools are combined into one system with performance and safety checks.
PSO3: Innovation in emerging areas with ethics and societal needs	3	Deep learning and computer vision are applied to a healthcare need, with ethical rules.

6.5 Mapping with SDGs

The project contributes to the United Nations Sustainable Development Goals (SDGs) listed in Table 6.4.

Table 6.4: Mapping of the project with SDGs

SDG	Contribution of the project
SDG 3: Good Health and Well-being	Helps stroke survivors to do their exercises correctly at home and gives an objective record of recovery to the therapist.
SDG 9: Industry, Innovation and Infrastructure	Applies computer vision and deep learning to build a low-cost rehabilitation tool using free software and ordinary hardware.
SDG 10: Reduced Inequalities	Reduces the cost and travel needed for rehabilitation, so patients who live far from a centre or have limited means are better served.

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APPENDIX

The appendix gives reference code sketches for the main modules of the system. The sketches in Appendix B (feature extraction and scoring) and Appendix D (database schema) were run on test data; the pose-extraction and model sketches in Appendices A and C show the structure of the modules and will be completed and tested in Semester VIII. They are not the final code.

A. Reference sketch: pose extraction with MediaPipe

The function reads a video, removes noise, finds the upper-body landmarks of every frame and returns an array of shape (frames, joints, 2), with missing values where the confidence is low.

```
# Reference sketch: keypoint extraction with MediaPipe + OpenCV
import cv2
import mediapipe as mp
import numpy as np

mp_pose = mp.solutions.pose
LM = mp_pose.PoseLandmark
USED = [LM.LEFT_SHOULDER, LM.RIGHT_SHOULDER, LM.LEFT_ELBOW,
        LM.RIGHT_ELBOW, LM.LEFT_WRIST, LM.RIGHT_WRIST,
        LM.LEFT_HIP, LM.RIGHT_HIP]

def extract_keypoints(video_path, min_conf=0.5):
    cap = cv2.VideoCapture(video_path)
    fps = cap.get(cv2.CAP_PROP_FPS) or 30
    frames = []
    with mp_pose.Pose(model_complexity=1) as pose:
        while True:
            ok, frame = cap.read()
            if not ok:
                break
            frame = cv2.GaussianBlur(frame, (3, 3), 0)   # noise removal
            res = pose.process(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
            if res.pose_landmarks is None:
                frames.append([[np.nan, np.nan]] * len(USED))
                continue
            lm = res.pose_landmarks.landmark
            frames.append([[lm[i].x, lm[i].y] if lm[i].visibility > min_conf
                           else [np.nan, np.nan] for i in USED])
    cap.release()
    return np.array(frames), fps      # shape (T, joints, 2)
```

B. Reference sketch: features, repetition counting and score

The code computes the joint angle (Equation (3.2)), the normalisation (Equation (3.1)), the smoothing, the speed, the repetition count and the weighted score of Table 3.3. The test at the end uses a synthetic angle signal with three repetitions; it printed `90.0` for the right-angle test, a count of 3 with a range of motion of 80°, and a score of 87.6 for the sample sub-scores.

```
import numpy as np
from scipy.signal import savgol_filter, find_peaks

def joint_angle(a, b, c):
    """Angle at point b (degrees) formed by points a-b-c (2-D or 3-D)."""
    a, b, c = map(np.asarray, (a, b, c))
    v1, v2 = a - b, c - b
    cosang = np.dot(v1, v2) / (np.linalg.norm(v1) * np.linalg.norm(v2) + 1e-9)
    return float(np.degrees(np.arccos(np.clip(cosang, -1.0, 1.0))))

def normalise(points, neck, l_sh, r_sh):
    """Express points relative to the neck, scaled by shoulder width."""
    scale = np.linalg.norm(np.asarray(l_sh) - np.asarray(r_sh)) + 1e-9
    return (np.asarray(points) - np.asarray(neck)) / scale

def smooth(signal, window=9, order=2):
    return savgol_filter(signal, window_length=window, polyorder=order)

def speed(track, fps=30):
    """Speed of a joint from its (T, 2) normalised track."""
    d = np.linalg.norm(np.diff(track, axis=0), axis=1)
    return d * fps

def count_repetitions(angle, fps=30, min_prom=15.0, min_gap_s=0.8):
    peaks, _ = find_peaks(angle, prominence=min_prom,
                          distance=int(min_gap_s * fps))
    return len(peaks), peaks

def score(rom, rom_target, smooth_s, speed_s, sym_s, model_p,
          w=(0.35, 0.20, 0.15, 0.10, 0.20)):
    s1 = min(rom / rom_target, 1.0)
    subs = (s1, smooth_s, speed_s, sym_s, model_p)
    return 100.0 * sum(wi * si for wi, si in zip(w, subs))

if __name__ == "__main__":
    print(joint_angle((0, 1), (0, 0), (1, 0)))      # 90.0
    t = np.linspace(0, 6, 180)                      # 6 s at 30 fps
    ang = 90 + 40 * np.sin(2 * np.pi * t / 2.0)     # 3 repetitions
    ang_s = smooth(ang)
    n, _ = count_repetitions(ang_s)
    print(n, round(ang_s.max() - ang_s.min(), 1))
    print(round(score(70, 80, 0.8, 1.0, 0.9, 0.85), 1))
```

C. Reference sketch: LSTM and hybrid models

```
# Reference sketch: LSTM / hybrid classifier (Keras)
```

```
from tensorflow import keras
from tensorflow.keras import layers

def build_lstm(T=60, F=20, n_classes=3):
    m = keras.Sequential([
        layers.Input(shape=(T, F)),
        layers.LSTM(64, return_sequences=True),
        layers.LSTM(64),
        layers.Dropout(0.3),
        layers.Dense(32, activation="relu"),
        layers.Dense(n_classes, activation="softmax"),
    ])
    m.compile(optimizer=keras.optimizers.Adam(1e-3),
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
    return m

def build_hybrid(T=60, F=20, n_classes=3):
    m = keras.Sequential([
        layers.Input(shape=(T, F)),
        layers.Conv1D(64, 5, padding="same", activation="relu"),
        layers.MaxPooling1D(2),
        layers.LSTM(64),
        layers.Dropout(0.3),
        layers.Dense(32, activation="relu"),
        layers.Dense(n_classes, activation="softmax"),
    ])
    m.compile(optimizer=keras.optimizers.Adam(1e-3),
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
    return m
```

D. Database schema

The three main tables in SQL (SQLite syntax). The tables *feedback_log* and *report* follow the same pattern.

```
CREATE TABLE patient (
    patient_id    INTEGER PRIMARY KEY,
    code          TEXT NOT NULL,
    age           INTEGER,
    affected_side TEXT CHECK (affected_side IN ('L','R')),
    therapist_id  INTEGER,
    start_date    TEXT
);
CREATE TABLE exercise (
    exercise_id   INTEGER PRIMARY KEY,
    name          TEXT NOT NULL,
    joint_triplet TEXT NOT NULL,
    target_rom    REAL,
    target_reps   INTEGER
);
CREATE TABLE session (
    session_id    INTEGER PRIMARY KEY,
    patient_id    INTEGER REFERENCES patient(patient_id),
    exercise_id   INTEGER REFERENCES exercise(exercise_id),
```

```
    date_time     TEXT,
    repetitions   INTEGER,
    mean_score    REAL,
    max_rom       REAL,
    mean_speed    REAL,
    smoothness    REAL,
    duration_s    REAL
);
```



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